# Project:Figure out IR from lammps for a CO2 molecule g16 and psi4

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#### Graphene + $CO_2$ System

### System Overview

- **System**: Graphene sheet + *n* $CO_2$ molecules
- **Variable**: *n* is determined based on pressure and temperature conditions.

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### $CO_2$ Structure Optimization

#### Gaussian Input File for Optimization

CO2_opt.gjf

```
%nprocshared=8
%mem=4GB
#p opt=tight b3lyp/6-311+g(d,p)

Title Card Required

0 1
 C                  0.34636872    0.34078212    0.00000000
 O                  1.60476872    0.34078212    0.00000000
 O                 -0.91203128    0.34078212    0.00000000
```

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### Morse Potential for Bond Parameter Fitting

#### Gaussian Input File

bond_scan.gjf

```
%nprocshared=8
%mem=4GB
#p opt=modredundant b3lyp/6-311+g(d,p)

Title Card Required

0 1
 C                 -0.65921789    0.93296088    0.00000000
 O                  0.18538211    0.93296088    0.00000000
 O                 -1.50381789    0.93296088    0.00000000
```

```
B 3 2 S 750 0.010000
```

**Morse Potential Fit Values**

- **De** = 2221.05716 kJ/mol (For 2 C=O bonds)
- **a** = 2.177624 (Actual used is 2)
- **re** = 1.1607 A

**Morse Potential Equation**

$$V(r) = D_e \left[1 - e^{-a(r-r_e)}\right]^2$$

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## Harmonic Potential for Angle Parameter Fitting

**Gaussian Input File**

angle_scan.gjf

```
%nprocshared=8
%mem=4GB
#p opt=modredundant b3lyp/6-311+g(d,p)

Title Card Required

0 1
 C                  0.00000000    0.00000000    0.00000000
 O                  0.00000000    0.74611250    0.88918226
 O                  0.00000000    0.74611250   -0.88918226

B 1 2 F
B 3 1 F
A 3 1 2 S 79 1.000000
```

**Angle Potential Fit Values**

- Data obtained from this input file was mirrored about the 180° value (since $CO_2$ bending is symmetric).
- **Parabolic Potential Fit Value**:
  - **a** = 0.07179377 kJ/mol.deg²
  - **theta_0** = 180 deg

**Parabolic Potential Equation**

$$V(\theta) = a(\theta - \theta_0)^2$$

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### ESP Calculation for RESP Charges

#### Gaussian Input File for ESP Calculation

CO2_esp.gjf

```
%nprocshared=8
%mem=4GB
#p b3lyp/6-311+g(d,p) pop=mk scf=tight iop(6/33=2)

Title Card Required

0 1
 C                  0.00000000    0.00000000    0.00000000
 O                  0.00000000    0.00000000    1.16079200
 O                  0.00000000    0.00000000   -1.16079200
```

#### Generating RESP Charges with Antechamber

Command used to obtain RESP charges from Antechamber:

```
antechamber -i CO2_esp.out -fi gout -o CO2_resp.mol2 -fo mol2 -c resp -nc 0
```

- The RESP charges are present inside the `ANTECHAMBER_RESP1.OUT` file.

### Obtained RESP Charges

| Atom | Charge |
|---|---|
| **C** | +0.75957 |
| **O** | -0.379785 |
| **O** | -0.379785 |

These RESP charges are only relavant in the Drude model.

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### Dispersion parameters

Since gaussian is unable to separate dispersion effects, we use SAPT0 available in PSI4 to calculate the dispersion energies for C-C , C-O and O-O interactions. The data is fitted to a buckingham potential.

## Dispersion Parameters for CO2 (from SAPT0)

| Interaction | A (kJ/mol) | ρ (Å⁻¹) | C (kJ/mol·Å⁶) |
|---|---|---|---|
| C–C | 1971.0064 | 0.3750 | 0 |
| C–O | 2352.8908 | 0.3286 | 0 |
| O–O | 3966.4545 | 0.2697 | 0 |

### Buckingham Potential Formula

$$V(r) = A\,e^{-r/\rho} - \frac{C}{r^6}$$

where:

- A = repulsion prefactor (eV),
- rho = inverse decay length (Å⁻¹),
- C = dispersion coefficient (eV·Å⁶),
- r = interatomic distance (Å).

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## Polarisable model 1 - QEq model

- Suitable for Reaxff potentials.
- Pressure dependent.
- Good where charge transfer occurs.

**LAMMPS input for correct IR at 1 atm using QEq model.**

IR.qeq.file3.works.lmp

```
units metal
dimension 3
boundary p p p
atom_style full

read_data CO2_atm.data

# Force fields with your Buckingham parameters (A in eV, ρ in Å⁻¹, C=0)
pair_style      buck/coul/long 14.0
pair_modify     tail yes
kspace_style    ewald 1.0e-4

# Buckingham coefficients (A, ρ, C=0)
pair_coeff 2 2 40.92922908874  0.269706714197472 0  # O-O
pair_coeff 1 1 20.33851015631 0.37501155251736 0 # C-C
pair_coeff 1 2 24.27911541566 0.269706714197472 0 # C-O
```

```
# Morse bond for CO₂ (keep your original values)
bond_style      morse
bond_coeff      1 11.45937 2 1.1607    # C-O bond

# Harmonic angle for CO₂ (linear)
angle_style     harmonic
angle_coeff     1 2.2 180.0              # O-C-O angle

# Initial charges (QEq will override these)
set            type 2 charge -0.379785        # O
set            type 1 charge 0.75957          # C

special_bonds lj/coul 0.0 0.0 0.5

fix 1 all qeq/shielded 1 1 1.0e-6 200 qeq.file3.txt

# Neighbor settings
neighbor        15.0 bin
neigh_modify    delay 0 every 1 check yes
# ----------------------------------------------------------------
# Initial minimization and equilibration
min_style cg
minimize 1.0e-10 1.0e-10 1000 10000
velocity all create 300.0 4928459 rot yes dist gaussian

#---------------------------------------------------------------
# Compute temperature and pressure
compute myTemp all temp
compute myPress all pressure myTemp

# Set output frequency (must match for thermo and fixes)
variable out_freq equal 100

# CUMULATIVE AVERAGING (strictly cumulative, updates every out_freq steps)
fix ave_temp all ave/time 1 ${out_freq} ${out_freq} c_myTemp ave running
fix ave_press all ave/time 1 ${out_freq} ${out_freq} c_myPress ave running

# Thermodynamic output (now includes cumulative averages)
thermo_style custom step temp press f_ave_temp f_ave_press vol density
thermo_modify flush yes
thermo ${out_freq}
```

```
# Equilibration in NVT
fix nvt_equil all nvt temp 300.0 300.0 100.0
timestep 0.0005
#run 200000
unfix ave_temp
unfix ave_press
reset_timestep 0
# ----------------------------------------------------------------
# IR Spectrum Calculation
# WARNING: System must have 0 net charge for dipole calculation
compute 1 all property/atom q xu yu zu
variable dipolex atom c_1[1]*c_1[2]
variable dipoley atom c_1[1]*c_1[3]
variable dipolez atom c_1[1]*c_1[4]
compute dxtot all reduce sum v_dipolex
compute dytot all reduce sum v_dipoley
compute dztot all reduce sum v_dipolez
variable totaldipolex equal c_dxtot
variable totaldipoley equal c_dytot
variable totaldipolez equal c_dztot
# Print dipole moment components with proper formatting
variable mytime equal step*dt
#fix printdipole all ave/time 1 1 2 v_mytime v_totaldipolex v_totaldipoley
v_totaldipolez file dipole_qeq.txt mode scalar format %.10f
#fix printdipole all print 1 v_mytime v_totaldipolex v_totaldipoley
v_totaldipolez file dipole.txt mode scalar format "%.3f %10.10f %10.10f
%10.10f" timestep no
fix printdipole all print 1 """$(v_mytime)  $(v_totaldipolex)
$(v_totaldipoley)  $(v_totaldipolez) """ title "" file dipole_qeq.txt screen
no

reset_timestep 0
dump 1 all custom 100 dump.co2.qeq.lammpstrj id type x y z q
thermo_style custom step temp press
run 200000
```

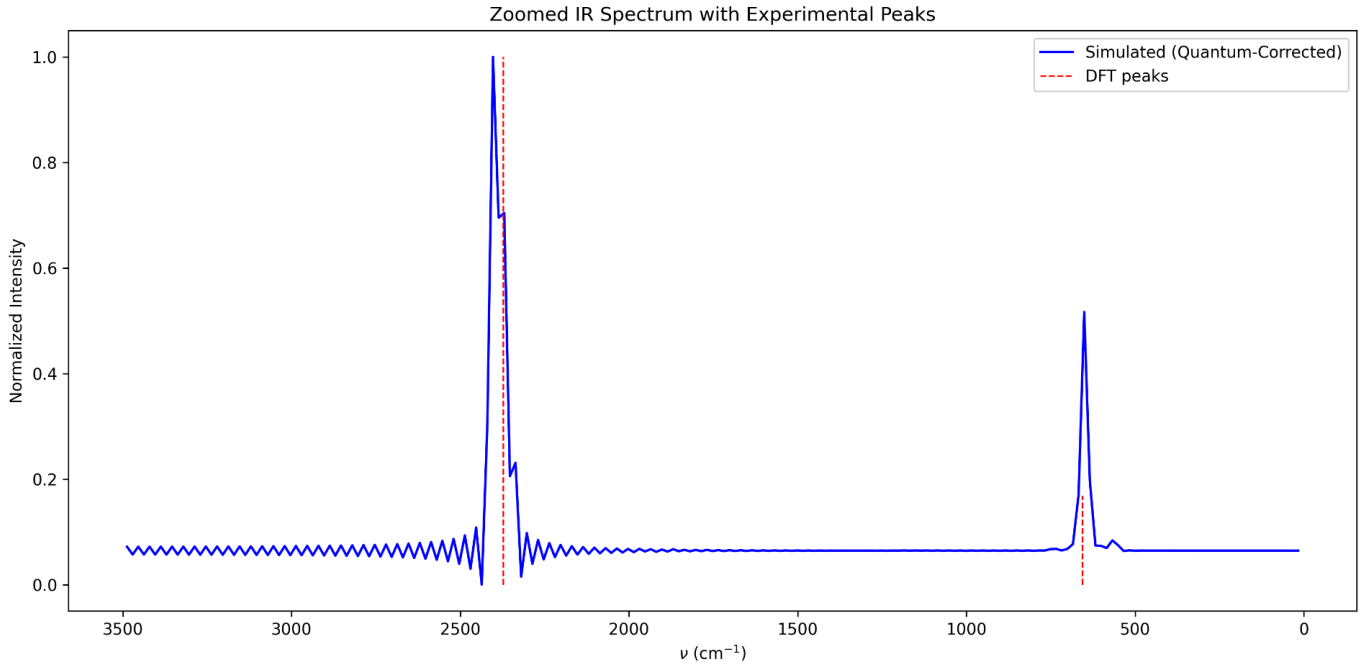
**Contents of qeq.file3.txt**

qeq.file3.txt

```
# Atom   χ(eV)   η(eV)   γ(Å⁻¹)   ζ(a.u.⁻¹)  q_core
1       4.3000    7.0     0.87     1.625      4     # Carbon
2      12.5000    9.5     0.87     2.275      6     # Oxygen
```

These values are fitted by trial-and-error to give correct charges of approximately +0.7 on carbon and -0.35 on oxygen.

**IR obtained from LAMMPS from this input file.**




*This system is a 50A x 50A x 450A with 27 CO2 molecules at 300K.*
*Uses simplified quantum correction of omega^3 proposed by Morita et al.*
*[Quantum corrections in vibrational and electronic condensed phase spectroscopy: Line shapes and echoes]*

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## Polarisable model 2 - Drude Oscillator.

- Most accurate for IR.
- Only considers polarisation, not applicable where charge transfer occurs.
  $q_{total} = q_{core} + q_{drude}$
  $q_{total} = RESPcharges$
  We need to specify the spring constant foe the charge shell of move.
  $k = q_{drude}^2/2 * \alpha_{site}$
  $\alpha_{site} = isotropic\ polarizability\ per\ site$

**Gaussian input file to calculate isotropic polarizability**
CO2_polar.gjf

```
%nprocshared=8
%mem=4GB
#p b3lyp/6-311+g(d,p) polar

Title Card Required

0 1
```

```
C                   0.00000000    0.00000000    0.00000000
O                   0.00000000    0.00000000    1.16079200
O                   0.00000000    0.00000000   -1.16079200
```

**Drude file with polarizabilties**

drude2.dff

```
# units: kJ/mol, A, deg
# type   m_D/u   q_D/e    k_D     alpha/A3   thole
C       0.2     -0.4     0.0      1.50       2.6
O       0.2      0.4     0.0      0.80       2.6
```

This file is used with the polarizer.py tool provided with the LAMMPS code to get the DC and DP edited in the CO2_atm.data file.

The values of the half-spring constant are calculated using the polarizabilities in the " drude.dff " file.

**LAMMPS input file for IR with drude**

IR.drude.CO2_atm2.data.works.lmp

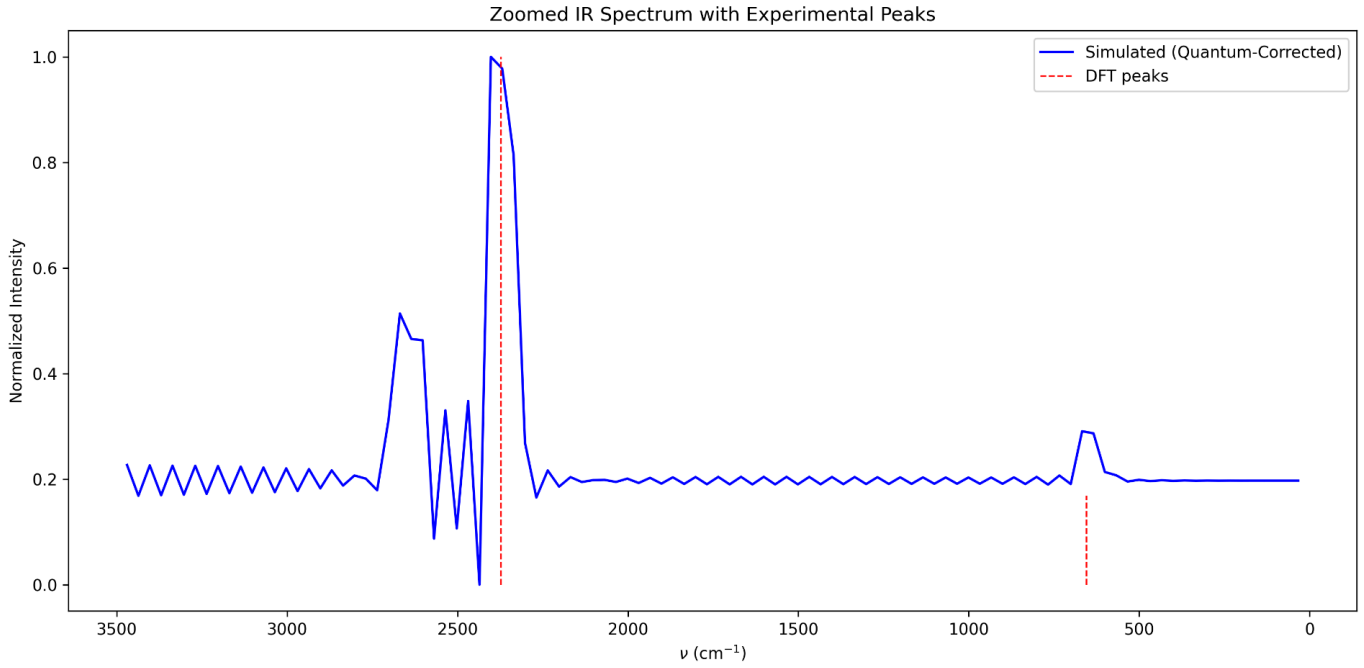
```
units metal
dimension 3
boundary p p p
atom_style full

# Read the atomic configuration
read_data CO2_atm2.data

# Charge setup (YOUR VALUES)
set type 1 charge 1.25957    # C core (+0.75957 - (-0.4))
set type 2 charge -0.879785  # O core (-0.379785 - 0.4)
set type 3 charge -0.5       # C Drude
set type 4 charge 0.5        # O Drude

# Define bond style and bond coefficients
bond_style hybrid morse harmonic
bond_coeff 1 morse 11.45937 2.0 1.1607    # Morse bond for C–O
bond_coeff 2 harmonic 1.02265625 0.00         # Harmonic bond for C–Drude
bond_coeff 3 harmonic 2.24984375 0.00         # Harmonic bond for O–Drude

# Define angle style and angle coefficients
angle_style harmonic
angle_coeff 1 2.2 180.0   # O–C–O angle
```

```
# Define pair style: combine Buckingham, Drude spring, and Thole damping
pair_style hybrid/overlay buck/coul/long 12.0 thole 2.6 12.0 coul/long/cs
12.0
pair_modify tail yes
kspace_style pppm 1.0e-4  # Use the CS-compatible Kspace solver

# Core-core Buckingham + Coulomb interactions
pair_coeff 1 1 buck/coul/long 20.33851015631 0.37501155251736 0     # C–C
pair_coeff 2 2 buck/coul/long 40.92922908874 0.269706714197472 0     # O–O
pair_coeff 1 2 buck/coul/long 24.27911541566 0.269706714197472 0     # C–O

# Drude spring electrostatics using coul/long/cs (Drude-core and Drude-
Drude)
pair_coeff * 3 coul/long/cs
pair_coeff * 4 coul/long/cs

# Thole damping for polarization
pair_coeff    1    1 thole   1.500
pair_coeff    1    2 thole   1.095
pair_coeff    1    3 thole   1.500
pair_coeff    1    4 thole   1.095
pair_coeff    2    2 thole   0.800
pair_coeff    2    3 thole   1.095
pair_coeff    2    4 thole   0.800
pair_coeff    3    3 thole   1.500
pair_coeff    3    4 thole   1.095
pair_coeff    4    4 thole   0.800

# Special bonds settings (Coulomb interactions for cores and Drude
particles)
special_bonds lj/coul 0.0 0.0 0.5

neighbor        15.0 bin
neigh_modify delay 0 every 1 check yes



# Define groups for core and Drude particles
group cores type 1 2      # Core atoms (C and O)
group drudes type 3 4     # Drude particles

# Apply the Drude oscillator model
fix DRUDE all drude 1 1 2 2
compute TDRUDE all temp/drude
```

```
compute TCOM cores temp/com

comm_modify vel yes

# Minimization and initial equilibration
min_style cg
minimize 1.0e-10 1.0e-10 1000 10000

variable out_freq equal 100
thermo_style custom step cpu etotal temp  press vol c_TDRUDE[1] c_TDRUDE[2]
c_TCOM
thermo 50
thermo_modify flush yes
thermo ${out_freq}

# Thermodynamic output
compute myTemp all temp
compute myPress all pressure myTemp

velocity cores create 300.0 4928459
velocity drudes create 1.0 4928459



timestep 0.0001
reset_timestep 0

# IR Spectrum Calculation (using Drude dipole moment)
compute 1 all property/atom q xu yu zu
variable dipolex atom c_1[1]*c_1[2]
variable dipoley atom c_1[1]*c_1[3]
variable dipolez atom c_1[1]*c_1[4]
compute dxtot all reduce sum v_dipolex
compute dytot all reduce sum v_dipoley
compute dztot all reduce sum v_dipolez
variable totaldipolex equal c_dxtot
variable totaldipoley equal c_dytot
variable totaldipolez equal c_dztot

# Print dipole moment components with proper formatting
variable mytime equal step*dt
fix printdipole all print 1 """$(v_mytime)  $(v_totaldipolex)
$(v_totaldipoley)  $(v_totaldipolez) """ title "" file dipole_drude.txt
screen no
```

```
# Output configurations and data
dump 1 all custom 100 dump.co2.drude.lammpstrj id type x y z q

fix fMOMENTUM all momentum 100 linear 1 1 1
fix tgnvt all tgnvt/drude temp 300.0 300.0 $(100.0*dt) 1.0 $(20.0*dt)
run 1000000
```

**IR obtained from LAMMPS from this input file.**

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## Static model

- RESP charges used
- Non-polarisable model

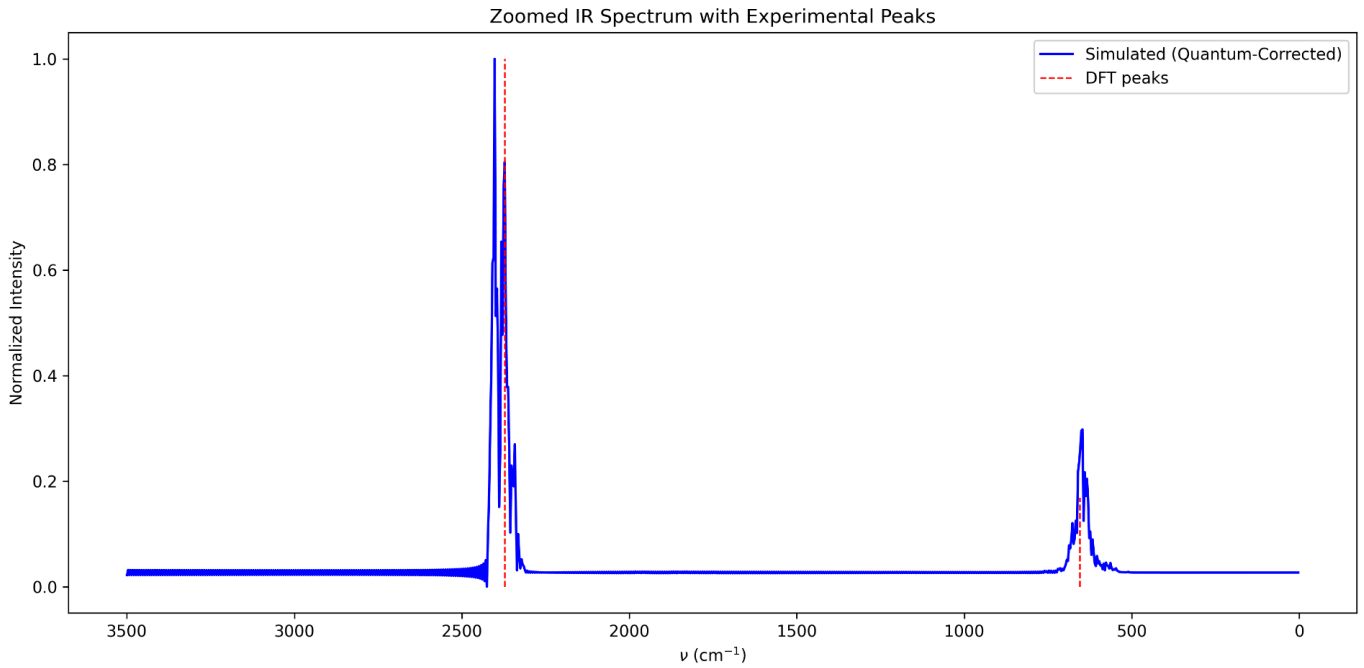
**LAMMPS input file for IR with static charges**
IR.stat.lmp

```
units metal
dimension 3
boundary p p p
atom_style full

# Read the atomic configuration
read_data CO2_atm.data

# Charge setup — constant charges (no Drude)
set type 1 charge 0.75957    # C core only
set type 2 charge -0.379785  # O core only
```

```
# Define bond style and bond coefficients
bond_style morse
bond_coeff 1 11.45937 2.0 1.1607    # Morse bond for C–O

# Define angle style and angle coefficients
angle_style harmonic
angle_coeff 1 2.2 180.0   # O–C–O angle

# Define pair style: Buckingham + long-range Coulomb
pair_style buck/coul/long 12.0
pair_modify tail yes
kspace_style pppm 1.0e-4

# Core-core Buckingham + Coulomb interactions
pair_coeff 1 1 20.33851015631 0.37501155251736 0     # C–C
pair_coeff 2 2 40.92922908874 0.269706714197472 0    # O–O
pair_coeff 1 2 24.27911541566 0.269706714197472 0    # C–O

# Special bonds (unchanged)
special_bonds lj/coul 0.0 0.0 0.5

# Neighbor settings
neighbor 15.0 bin
neigh_modify delay 0 every 1 check yes

# Groups – only cores exist now
group cores type 1 2

# Communication settings
comm_modify vel yes

# Minimization
min_style cg
minimize 1.0e-10 1.0e-10 1000 10000

# Thermo output
variable out_freq equal 100
thermo_style custom step cpu etotal temp press vol
thermo ${out_freq}
thermo_modify flush yes

# Compute temperature/pressure
```

```
compute myTemp all temp
compute myPress all pressure myTemp

# Initial velocities — add rotational degrees of freedom
velocity cores create 300.0 4928459 rot yes dist gaussian

# Timestep
timestep 0.0005  # faster possible without Drude
reset_timestep 0

# IR Spectrum Calculation (dipole from charges)
compute dipole all property/atom q xu yu zu
variable dipolex atom c_dipole[1]*c_dipole[2]
variable dipoley atom c_dipole[1]*c_dipole[3]
variable dipolez atom c_dipole[1]*c_dipole[4]
compute dxtot all reduce sum v_dipolex
compute dytot all reduce sum v_dipoley
compute dztot all reduce sum v_dipolez

variable totaldipolex equal c_dxtot
variable totaldipoley equal c_dytot
variable totaldipolez equal c_dztot

# Time variable
variable mytime equal step*dt

# Dipole output
fix printdipole all print 1 "${mytime} ${totaldipolex} ${totaldipoley}
${totaldipolez}" title "" file dipole_stat.txt screen no

# Trajectory dump
dump 1 all custom 100 dump.co2.stat.lammpstrj id type x y z q

# Momentum fix
fix fMOMENTUM all momentum 100 linear 1 1 1

# Thermostat (regular NVT)
fix NVT all nvt temp 300.0 300.0 0.5

# Main run
run 1000000
```

**IR obtained from LAMMPS from this input file**




## Machine learning model

- orb-v3-conservative-inf-omat model used
- Since there is no concept of charges, a bond length vector in the direction of the dipole is considered as the dipole vector.

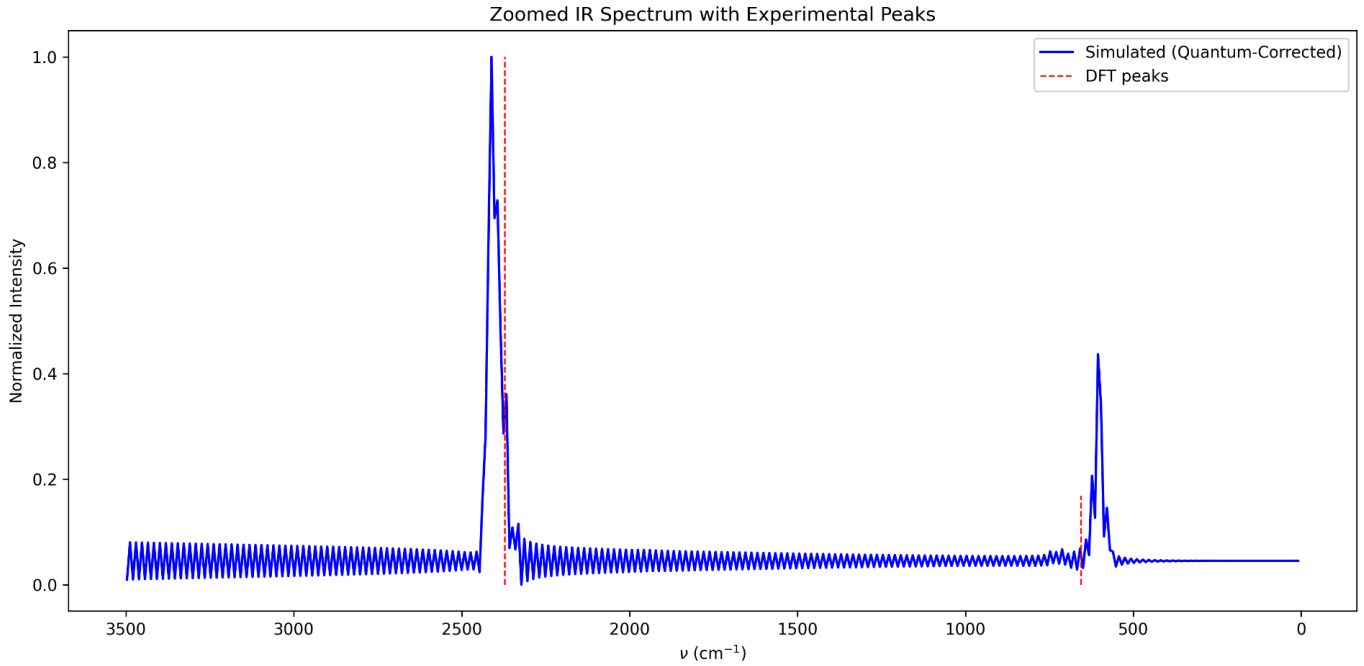
**LAMMPS input file for IR with orb-models**

IR.orb.lmp

```
units metal
dimension 3
boundary p p p
atom_style full

read_data CO2_atm.data

bond_style zero
bond_coeff 1

pair_style  gnnp/gpu gnnp_driver.py
pair_coeff  * *  orb orb-v3-conservative-inf-omat C O

# Neighbor
neighbor 15.0 bin
neigh_modify delay 0 every 1 check yes

# Groups
group CO2 type 1 2
```

```
# Minimization (optional)
min_style cg
minimize 1.0e-10 1.0e-10 1000 10000

# Velocity with rotation
velocity CO2 create 300.0 4928459 rot yes mom yes dist gaussian

# NVT thermostat
timestep 0.0005
fix NVT CO2 nvt temp 300.0 300.0 0.5

# --- Dipole vector calculation using bond vectors ---
# Get bond vectors: bond/local
compute B all bond/local dist dx dy dz

# Reduce dipole components: sum of C–O bond vectors
compute DXTOT all reduce sum c_B[2]
compute DYTOT all reduce sum c_B[3]
compute DZTOT all reduce sum c_B[4]

# Variables for printing
variable dipx equal c_DXTOT
variable dipy equal c_DYTOT
variable dipz equal c_DZTOT
variable mytime equal step*dt

# Output dipole vector vs time
fix printDIPOLE all print 1 """$(v_mytime)  $(v_dipx)  $(v_dipy)  $(v_dipz)
""" file dipole_orb.txt screen no

# Thermo
thermo_style custom step temp etotal press vol
thermo 100

# Dump
dump 1 all custom 100 dump.orb.lammpstrj id type x y z

# Run
run 400000
```

**IR obtained from LAMMPS from this input file**


Zoomed IR Spectrum with Experimental Peaks
Simulated (Quantum-Corrected)
DFT peaks
Normalized Intensity
1.0
0.8
0.6
0.4
0.2
0.0
3500
3000
2500
2000
1500
1000
500
0
ν (cm⁻¹)